Quiz 1

CS 102 Computer Programming I · Fall Semester 2026
Week 4 · Duration: **35 MINUTES** · Version **C**

New Uzbekistan University

FULL NAME: _____ ID: _____ GROUP: _____

Section	Points	Score
A Multiple choice	30	
B Fill in the gaps	40	
C Trace and write	30	
Total	**100**	

Rules. No notes, phones or computers. Write answers *only* in the space given. For fill-in-the-gap questions write exactly one piece of C++ code in each numbered gap — a keyword, an operator, a type or an expression. Assume every code fragment is inside `main` with `#include <iostream>` and `using namespace std;`.

Section A: Multiple Choice (30 points)

Circle **one** answer for each question.

A1. (6 points) The hexadecimal literal `0x1A` has the decimal value: A. 16 B. 20 C. 26 D. 110

A2. (6 points) What does `cout << 11 / 4 * 2.0 + 11 % 4;` print? A. 7 B. 8.5 C. 9.5 D. 6

A3. (6 points) What does this code print?

```
int x = 0;
if (x != 0 && 10 / x > 1) {
    cout << "big ";
}
cout << x + 1;
```

A. `big 1` B. `1` C. it crashes (division by zero) D. `0`

A4. (6 points) Which expression does **not** evaluate to `4.5`?

A. `9 / 2.0`

B. `static_cast<double>(9 / 2)`

C. `static_cast<double>(9) / 2`

D. `9.0 / 2`

A5. (6 points) What does this print? `char c = 'B' + 3; cout << c << (c - 1);`
Hint: in the ASCII table, `'A'` *= 65.* A. `ED` B. `E 68` C. `6968` D. `E68`

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Section B: Fill in the Gaps (40 points)

Each numbered gap is worth 2 points. Every program must compile and produce the output shown. Write your answers in the boxes under each program.

B1. (10 points) **Reverse a three-digit number.** Input `472` must print `274`; `305` prints `503`.

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cin (1)_____ n;
    int h = n (2)_____ 100;              // hundreds digit
    int t = n / 10 (3)_____ 10;          // tens digit
    int u = n % 10;                  // units digit
    int r = u (4)_____ 100 + t * 10 + h;
    cout << r << (5)_____;
    return 0;
}
```

(1) _____ (2) _____ (3) _____ (4) _____ (5) _____

B2. (10 points) **Average speed.** Distance in km and time in minutes are integers and must both be positive. `Too fast` above 120 km/h, `Normal` from 60, otherwise `Slow`. Input `150 90` must print `Speed: 100, rounded: 100, Normal`; `100 45` prints `Speed: 133.333, rounded: 133, Too fast`.

```
int dist, time;
cin >> dist >> time;
if ((6)_____) {           // reject bad input
    cout << "Invalid" << endl;
} else {
    double speed = (7)_____(dist) / time * 60;   // km per hour
    cout << "Speed: " << speed << ", rounded: " << (8)_____(speed + 0.5);
    if (speed (9)_____ 120) {
        cout << ", Too fast" << endl;
    } else if (speed (10)_____ 60) {
        cout << ", Normal" << endl;
    } else {
        cout << ", Slow" << endl;
    }
}
```

(6) _____ (7) _____ (8) _____ (9) _____ (10) _____

B3. (10 points) **Quadrant of a point.** Input `3 4` prints `Quadrant 1`; `-2 5` prints `Quadrant 2`; `0 7` prints `On an axis`; `4 -1` prints `Quadrant 4`.

```
int x, y;
cin >> x >> y;
if (x (11)_____ 0 (12)_____ y == 0) {
    cout << "On an axis" << endl;
} else if (x > 0 (13)_____ y > 0) {
    cout << "Quadrant 1" << endl;
} else if (x < 0 && y (14)_____ 0) {
    cout << "Quadrant 2" << endl;
} else if (x < 0 && y < 0) {
    cout << "Quadrant 3" << endl;
} (15)_____ {
    cout << "Quadrant 4" << endl;
}
```

(11) _____ (12) _____ (13) _____ (14) _____ (15) _____

B4. (10 points) **Field size (watch the area!).** Length and width are up to 10^5 metres. A field is *large* if its area is at least 2×10^9 m^2; it is *square* if length equals width. Input `50000 50000` must print `Both`.

```
int length, width;
cin >> length >> width;
(16) _____ area = static_cast<long long>(length) * width;
bool large = area >= 2000000000;
bool square = length == width;
if (large (17) _____ square) {
    cout << "Both" << endl;
} else if ((18) _____) {
    cout << "Large" << endl;
} else if (square) {
    cout << "Square" << endl;
} (19) _____ {
    cout << "Neither" << endl;
}
```

(16) [ ] (17) [ ] (18) [ ] (19) [ ]

(20) Why would `int area = length * width;` print `Square` for the input above?

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Section C: Trace and Write (30 points)

C1. (15 points) Write **exactly** what this fragment prints (mind the spaces).
Hint: in the ASCII table, 'a' = 97 (and 'A' = 65).

```
int a = 5, b = 8, c = 2;
double d = b / a + b % a / 2.0;
if (b > a)
    if (c > 5 || b / c > 5)
        cout << "P";
else
    cout << "Q";
cout << " " << d;
char ch = 'A' + c;
if (ch == 'C')
    ch = ch + 1;
cout << " " << ch << " " << (ch - 'A') << endl;
```

C2. (15 points) **Length of three lessons.** Write a program that reads the lengths of three lessons in minutes and prints their total length as hours and minutes. Use only `int`, `/` and `%`.

```
80 95 50   → 3 h 45 min
45 45 45   → 2 h 15 min
20 20 20   → 1 h 0 min
```

END OF QUIZ